Chapter 3 : Data Link Layer

3.1. Data link layer functions - Framing, error control, flow control, access control

The Data Link Layer is the second layer of the OSI model. It is responsible for reliable node-to-node delivery of data. It takes data from the Network Layer above, breaks it into frames, and sends these frames to the Physical Layer, ensuring that they arrive at the destination correctly.

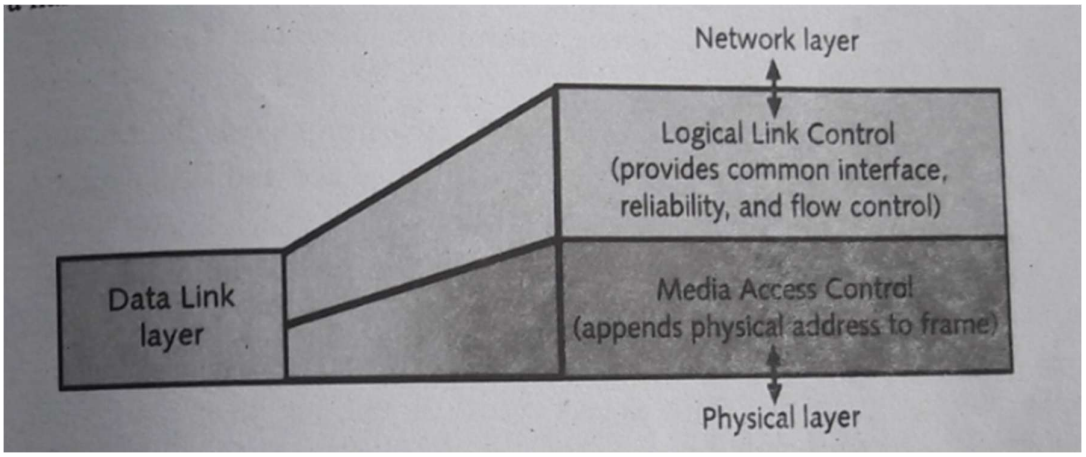


fig 1: sub layers of data link layer

The key functions of the Data Link Layer are:

1. Framing :

Framing is the process of dividing a stream of bits received from the Network Layer into manageable data units called frames. Each frame has a clearly defined beginning and end so the receiver can identify frame boundaries.

- Data is sent in blocks called frames
- The beginning and end of each frame must be recognizable

  Frames can be of fixed size or variable size

There are basically two different types of framing :

Fixed - Size Framing :

The frame is of a predetermined fixed size. No need to define boundaries - the length of the frame itself acts as a delimiter.

- Disadvantage: Suffers from internal fragmentation if the actual data is smaller than the frame size. Padding must be added.

Variable - Size Framing :

The frame size varies depending on data. The boundary between frames must be explicitly marked. Two methods are used:

- Length Field: A length field in the frame header indicates the size of the frame. Used in Ethernet (802.3).

  Problem: If the length field gets corrupted, framing breaks down.

- End Delimiter (ED): A special pattern (flag sequence) marks the end of a frame. Used in Token Ring.

  Problem: If the ED pattern appears inside the data, it creates ambiguity. Solved using byte stuffing or bit stuffing.

Character/Byte Stuffing :

Used when frames consist of character. If data contains ED, then, byte is stuffed into data to differentiate it from ED.

- Disadvantage – It is very costly and obsolete method.

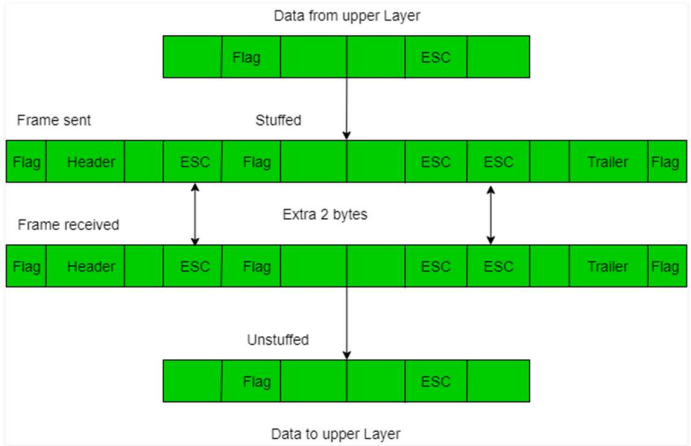


fig 2 : Character/Byte Stuffing

Bit Stuffing :

Let ED = 01111 and if data = 01111

–> Sender stuffs a bit to break the pattern i.e. here appends a 0 in data = 011101.

–> Receiver receives the frame.

–> If data contains 011101, receiver removes the 0 and reads the data.

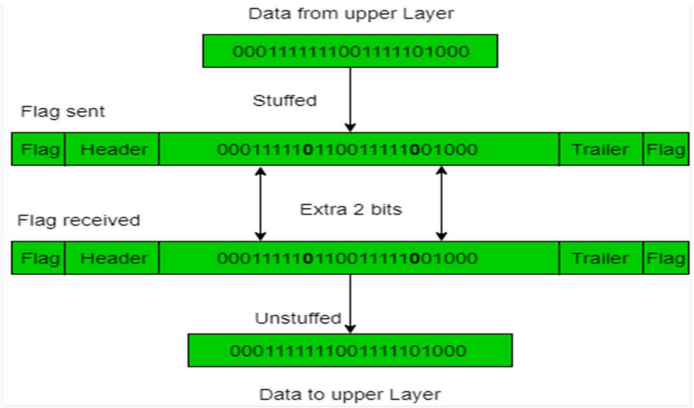


fig 3 : Bit Stuffing

2. Error Control :

Bit errors inevitably occur during transmission due to noise and interference. Error control refers to mechanisms to detect and/or correct these errors.

- Error Detection: Identify whether the received frame contains errors (e.g., Parity, Checksum, CRC).
- Error Correction: Correct the errors without retransmission (e.g., Hamming Code).
- ARQ (Automatic Repeat Request): Request the sender to retransmit when an error is detected.

3. Flow Control :

Flow control coordinates the amount of data that can be sent before the sender must wait for an acknowledgement from the receiver. Without flow control, a fast sender can overwhelm a slow receiver.

- The receiver has limited buffer memory and processing speed
- The receiver informs the sender before its buffer fills up
- Flow control ensures the sender does not transmit faster than the receiver can absorb

Main flow control mechanisms:

- Stop-and-Wait: Sender sends one frame and waits for ACK before sending the next.
- Sliding Window: Sender can send multiple frames (up to window size W) before needing ACK.

4. Access Control :

When multiple stations share a single communication link (broadcast channel), access control determines how stations take turns to transmit without interfering with each other. This is handled by the MAC sublayer.

- Contention-based: Stations compete for the channel (ALOHA, CSMA, CSMA/CD, CSMA/CA).
- Controlled Access: Access is controlled centrally or via tokens (Polling, Token Passing).
- Channelization: The channel is divided among stations (FDMA, TDMA, CDMA).

3.2. Error control codes: Parity, checksum, CRC

Errors in data transmission can be single-bit errors or burst errors. Error control codes add redundant bits to data to enable detection or correction of these errors.

Types of Errors :

- **Single-Bit Error:** Only one bit in the data unit is changed from 1 to 0 or from 0 to 1. Rare in serial data transmission.

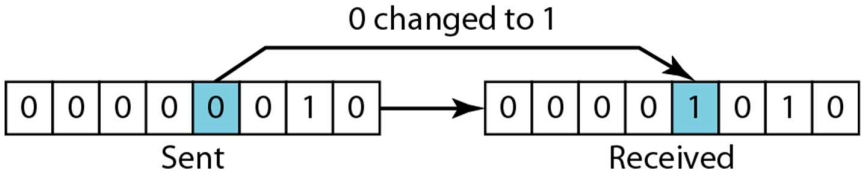


fig 4 : Single Bit error

- **Burst Error:** Two or more bits in the data unit have changed from 1 to 0 or 0 to 1. More common in practice due to the duration of noise relative to data rate.

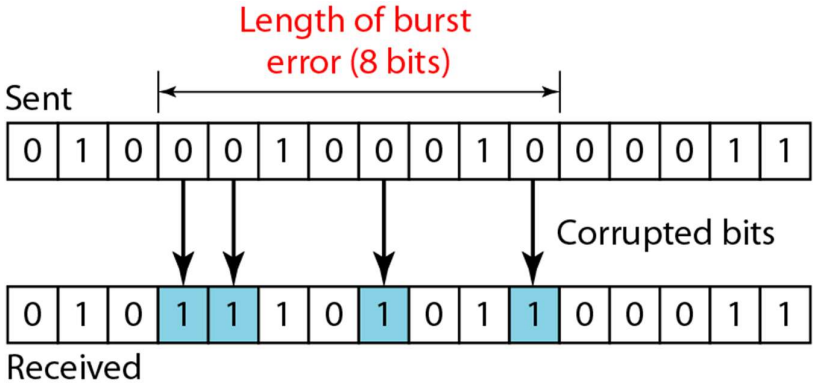


Fig 5: Burst error of length 8

Parity Check :

Parity is the simplest error detection method. A single parity bit is added to the data to make the total number of 1s either even (even parity) or odd (odd parity).

Even Parity :

- A parity bit is added such that the total number of 1 bits (including the parity bit) is even.
- Example: Data = 1011001. Number of 1s = 4 (even). Parity bit = 0. Transmitted: 10110010.

Odd Parity :

- A parity bit is added to make the total number of 1 bits odd.
- Example: Data = 1011001. Number of 1s = 4. Parity bit = 1. Transmitted: 10110011.
- Limitation: Parity can only detect an odd number of bit errors. If two bits flip, parity remains unchanged and the error goes undetected. It cannot detect burst errors reliably.

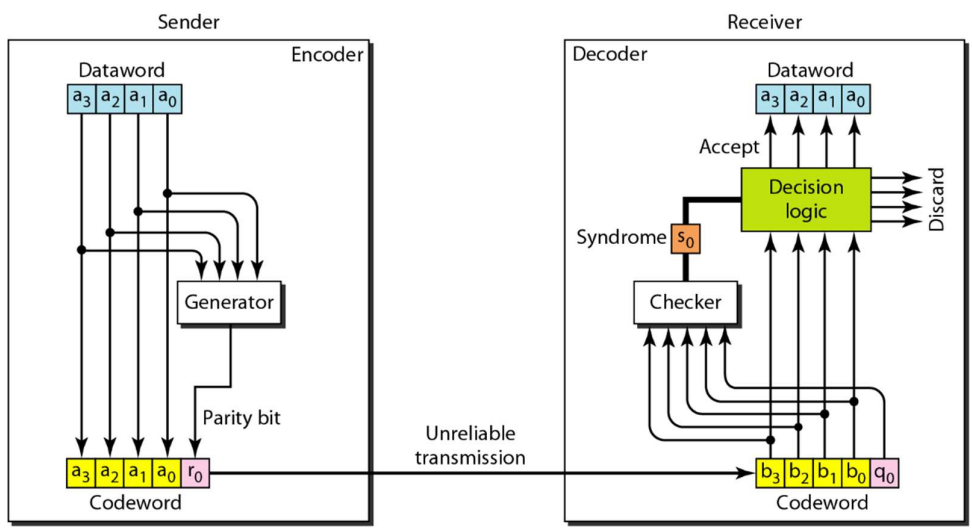


fig 6 : Simple parity check

Checksum :

Checksum is more robust than parity. It treats data as a sequence of integers and computes a sum used for error detection.

How Checksum Works (Sender) :

1) Divide the data into fixed-size segments (e.g., 16-bit segments).
2) Add all segments together using one's complement arithmetic.
3) Compute the one's complement of the sum — this is the checksum.
4) Append the checksum to the data and transmit.

Verification (Receiver) :

1) Add all received segments including the checksum using one's complement arithmetic.
2) If the result is all 1s (i.e., -0 in one's complement), no error is detected.
3) If any bit is 0, an error has occurred.

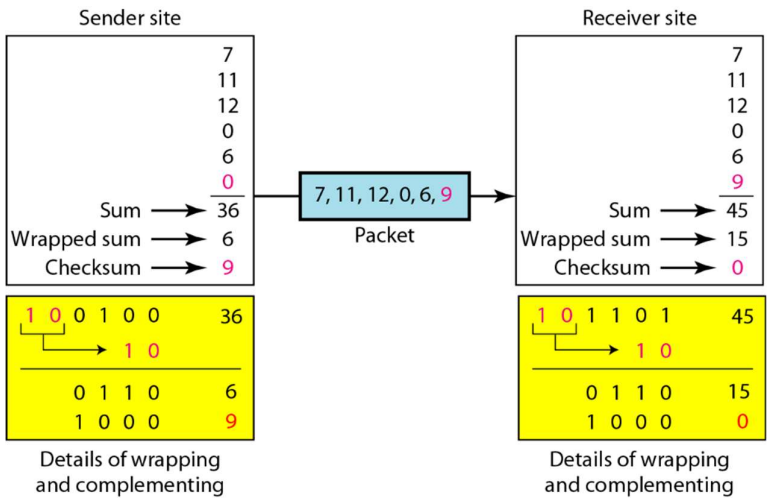


fig 7 : Checksum

Cyclic Redundancy Check :

CRC is the most powerful and widely used error detection method. It treats the data as a large binary polynomial and divides it by a predetermined generator polynomial. The remainder is the CRC code, which is appended to the data.

Both sender and receiver agree on a generator polynomial G(x). The sender appends (degree of G(x)) zero bits to the data, divides by G(x) using binary (mod-2) division, and appends the remainder as the CRC.

CRC Sender Steps :

1) Let the message M(x) and generator G(x) of degree r be agreed upon.
2) Append r zero bits to M(x) to get M'(x).
3) Divide M'(x) by G(x) using module-2 division (XOR division).

4) The remainder R(x) is the CRC. Append R(x) to the original message M(x).
5) Transmit the combined frame T(x) = M(x) + R(x).

CRC Receiver Steps :

1) Divide the received frame T(x) by G(x) using module-2 division.
2) If the remainder is 0, no error is detected.
3) If the remainder is non-zero, a transmission error has occurred.

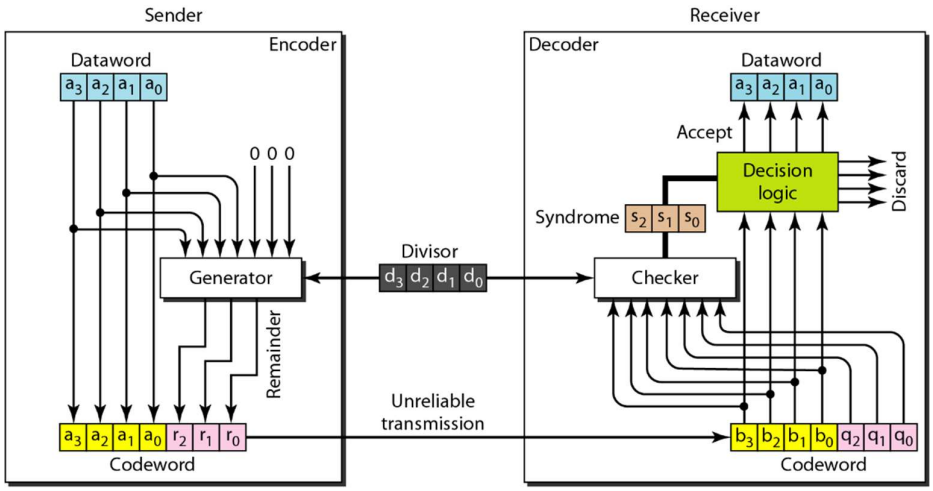


fig 8 : CRC encoder and decoder

Example :

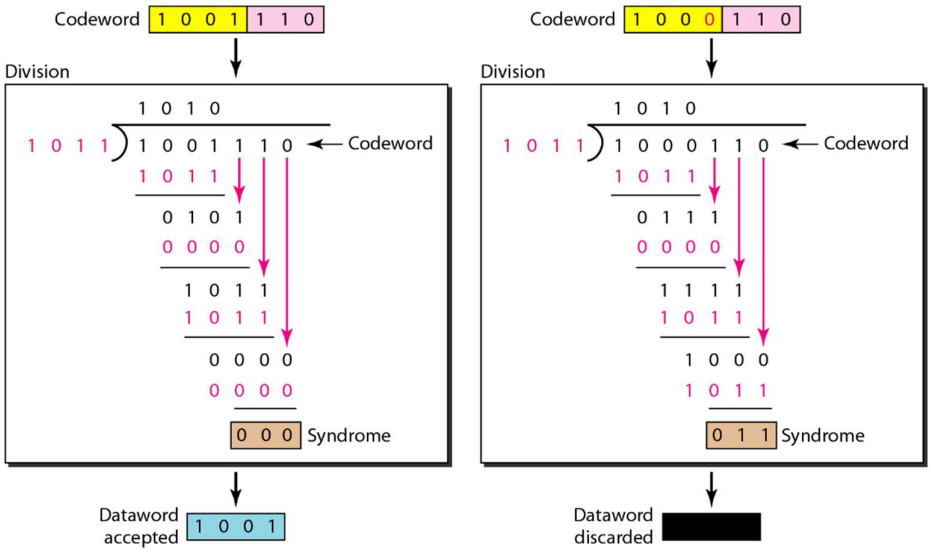


fig 9 : CRC division in two cases

Name	*Polynomial*	*Application*
CRC-8	$x^8 + x^2 + x + 1$	ATM header
CRC-10	$x^{10} + x^9 + x^5 + x^4 + x^2 + 1$	ATM AAL
CRC-16	$x^{16} + x^{12} + x^5 + 1$	HDLC
CRC-32	$x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x + 1$	LANs

fig 10 : Standard Polynomials

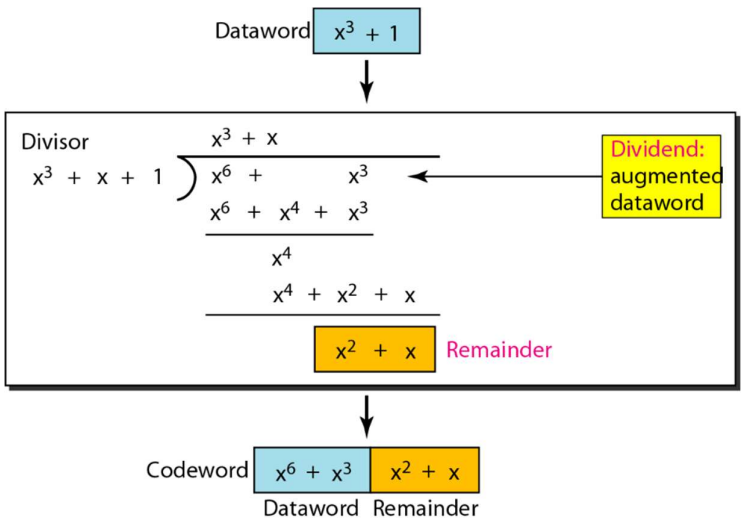


fig 11 : CRC division using polynomials

Hamming Code (Error Correction) :

Hamming code is a forward error correction (FEC) technique that not only detects but also corrects single-bit errors. It adds redundant parity bits at specific positions (powers of 2).

Rules for Hamming Code :

- Redundant bits are placed at positions that are powers of 2: position 1, 2, 4, 8, 16,…
- Each redundant bit is the parity bit for a specific group of data bits
- `Minimum number of parity bits r for m data bits: 2^r ≥ m + r + 1`

Example: Encode data 1011 using Even Parity Hamming Code

`Data bits m = 4. We need r parity bits where 2^r ≥ 4 + r + 1. For r = 3: 2^3 = 8 ≥ 8. So, 3 parity bits needed.`

- Total bits = m + r = 4 + 3 = 7 bits.
- Parity bits at positions: 1 (P1), 2 (P2), 4 (P4).
- Data bits at positions: 3 (d1=1), 5 (d2=0), 6 (d3=1), 7 (d4=1).

- P1 covers bits at positions 1, 3, 5, 7 → P1 makes even parity of {1,0,1} → P1 = 0
- P2 covers bits at positions 2, 3, 6, 7 → P2 makes even parity of {1,1,1} → P2 = 1
- P4 covers bits at positions 4, 5, 6, 7 → P4 makes even parity of {0,1,1} → P4 = 0
- Hamming code: P1 P2 d1 P4 d2 d3 d4 = 0 1 1 0 0 1 1 = 0110011

Error Detection and Correction :

1) At receiver, recalculate all parity checks.
2) The binary number formed by the failing parity checks gives the position of the erroneous bit.
3) Flip that bit to correct the error.

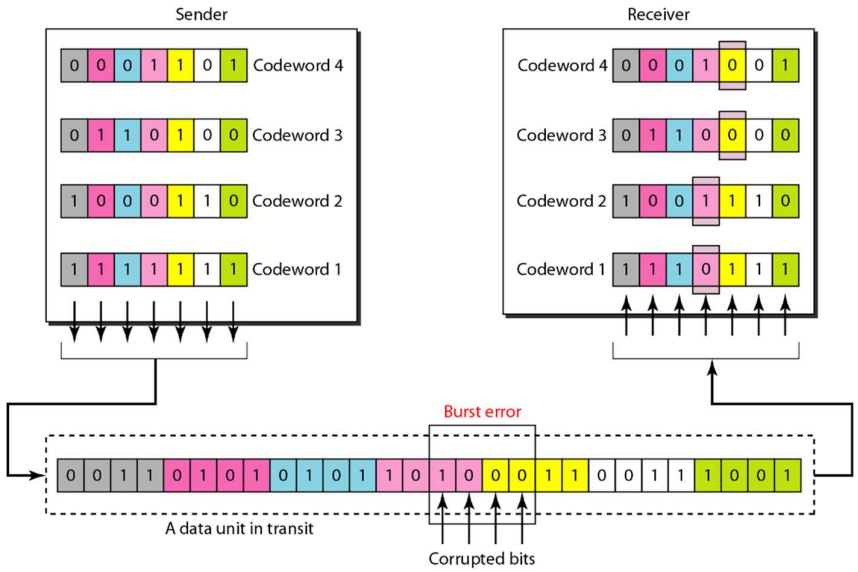


fig 12 : Burst Error using Hamming Code

Error and Flow Control Mechanisms :

- Stop-and-Wait
- Sliding Window
- Go-Back-N ARQ
- Selective-Repeat ARQ

1) Stop-and-Wait :

- A source transmits a frame
- After the destination entity receives the frame, it indicates its willingness to accept another frame by sending back an acknowledgement to the frame just received
- The source must wait until it receives the acknowledgement before sending the next frame
- Receiver has a control variable that holds the number of the next frame expected (0 or 1)
- The process of alternately sending and waiting repeats until the sender transmits the end of transmission(EoT) frame
- The destination can thus stop the flow of data simply by withholding acknowledgement
- This procedure is inherently a half-duplex because the transmission is in both directions but only in one direction at a time

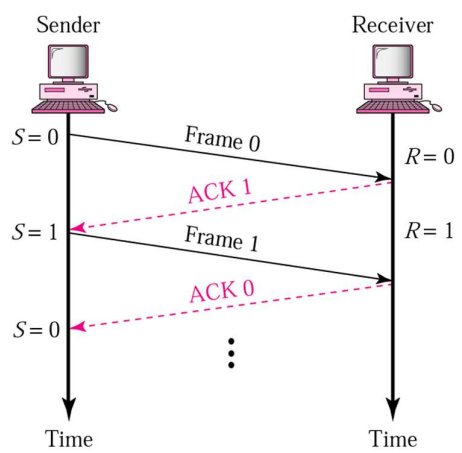


fig 13 :Stop and Wait Flow Control

Stop-And-Wait ARQ : LOST ACK FRAME :

- When a receiver receives a damaged frame, it discards it and keeps its value of R
- After the timer at the sender expires, another copy of frame 1 is sent
- If the sender receives a damaged ACK, it discards it
- When the timer of the sender expires, the sender retransmits frame 1

- Receiver has already received frame 1 and expecting to receive frame 0 (R=0). Therefore, it discards the second copy of frame 1

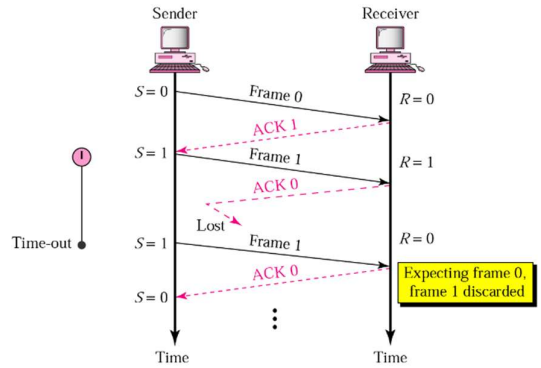


fig 14 : Stop and Wait ARQ

Stop-And-Wait ARQ: DELAYED ACK FRAME :

- The ACK can be delayed at the receiver or due to some problem
- It is received after the timer for frame 0 has expired
- Sender retransmitted a copy of frame 0. However, R =1 means the receiver expects to see frame 1. Receiver discards the duplicate frame 0
- Sender receives 2 ACKs, it discards the second ACK

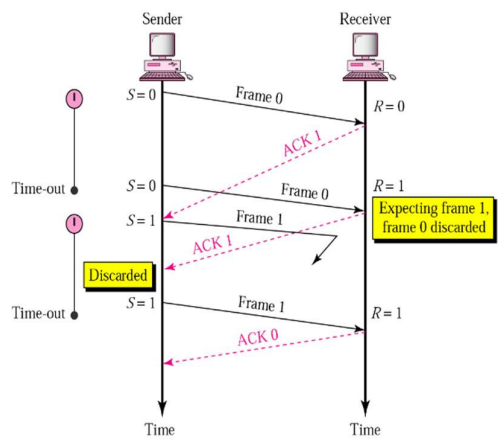


Fig 15 : Stop and Wair ARQ: Delayed ARQ Frame

2) Sliding Window :

Sender Sliding Window :

- At the sending site, to hold the outstanding frames until they are acknowledged, we use the concept of a window
- The size of the window is at most 2^m -1 where m is the number of bits for the sequence number

- Size of the window can be variable, e.g. TCP
- The window slides to include new unsent frames when the correct ACKs are received

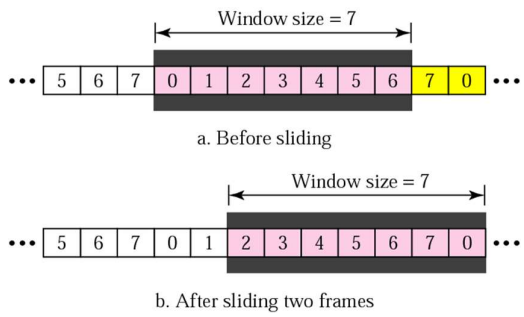


fig 16 : Sender Sliding Window

Receiver Sliding Window :

- Size of the window at the receiving site is always 1 in this protocol
- Receiver is always looking for a specific frame to arrive in a specific order
- Any frame arriving out of order is discarded and needs to be resent
- Receiver window slides as shown in fig. Receiver is waiting for frame 0 in part a

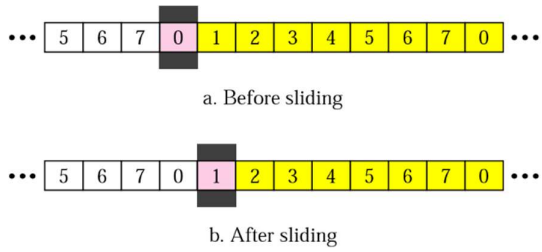


fig 17 : Receiver Sliding Window

3) Go-Back-N ARQ :

- The sender keeps track of the outstanding frames and updates the variables and windows as the ACKs arrive
- Frame 2 is lost
- When the receiver receives frame 3, it discards frame 3 as it is expecting frame 2 (according to window)
- After the timer for frame 2 expires at the sender site, the sender sends frame 2 and 3. (go back to 2)

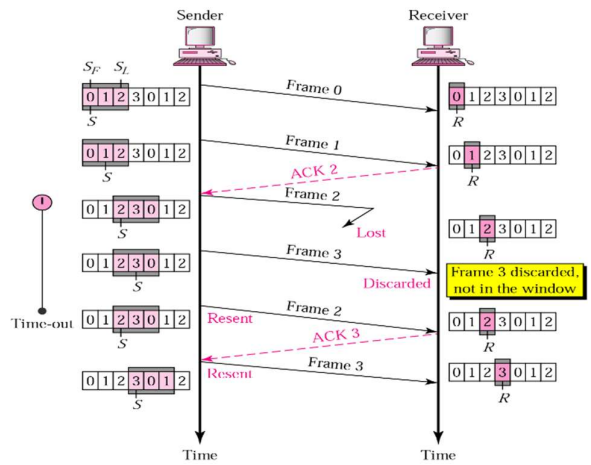
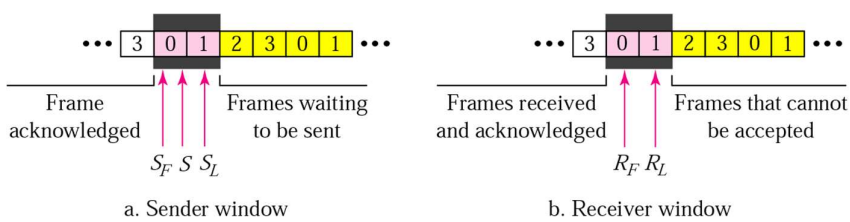


fig 18 : Go-Back-N ARQ

4) Selective-Repeat ARQ :

- Go-Back-N ARQ simplifies the process at the receiver site. Receiver only keeps track of only one variable, and there is no need to buffer out-of-order frames, they are simply discarded
- However, Go-Back-N ARQ protocol is inefficient for noisy link. Its bandwidth inefficient and slows down the transmission
- In Selective Repeat ARQ, only the damaged frame is resent. More bandwidth efficient but more complex processing at receiver
- It defines a negative ACK (NAK) to report the sequence number of a damaged frame before the timer expires



a. Sender window b. Receiver window

Fig 19 : Sender and Receiver Window of selective repeat ARQ

3.3. Data link protocols: PPP, HDLC

High-Level Data Link Control (HDLC) :

HDLC is a bit-oriented, synchronous data link protocol developed by ISO. It is used for both point-to-point and multipoint communications. Data is organized into frames, and each frame is verified upon arrival.

Transfer Modes in HDLC :

- Normal Response Mode (NRM): A primary station sends commands; secondary stations respond. Used for point-to-point and multipoint links. Communication is always initiated by the primary.
- Asynchronous Balanced Mode (ABM): Each station is a combined station and can both send commands and respond. Used only for point-to-point communications. This is the most used mode today (e.g., in PPP).

HDLC Data Frame :

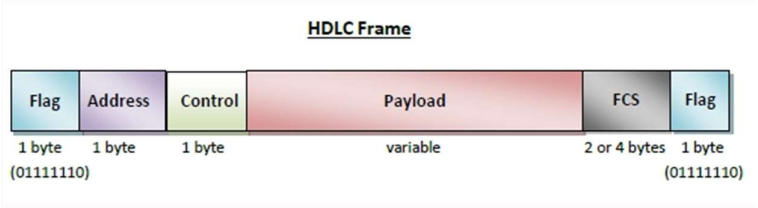


fig 20 : HDLC Data Frame

Field	Size	Description
Flag	1 byte (8 bits)	Marks start and end of frame. Bit pattern: 01111110
Address	1 byte	Address of sender or receiver (secondary station's address)
Control	1 byte	Sequence numbers, acknowledgements, and frame type info
Payload (Data)	Variable	Data received from the Network Layer
FCS	2 or 4 bytes	Frame Check Sequence using CRC for error detection
Flag	1 byte	End flag: 01111110

Types of HDLC Frames :

- **I-frame** – I-frames or Information frames carry user data from the network layer. They also include flow and error control information that is piggybacked on user data. The first bit of control field of I-frame is 0.
- **S-frame** – S-frames or Supervisory frames do not contain information field. They are used for flow and error control when piggybacking is not required. The first two bits of control field of S-frame is 10.
- **U-frame** – U-frames or Un-numbered frames are used for myriad miscellaneous functions, like link management. It may contain an information field, if required. The first two bits of control field of U-frame is 11.

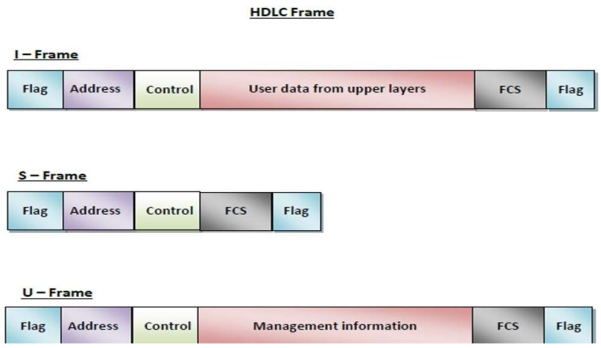


fig 21 : Types of HDLC frames

Point-to-Point Protocol (PPP) :

PPP is a byte-oriented data link protocol used to transmit multi-protocol data between two directly connected (point-to-point) computers. It is widely used for broadband Internet connections (DSL, dial-up). Also known as RFC 1661.

Services Provided by PPP :

- Defining the frame format for data transmission
- Establishing and terminating links between two endpoints
- Encapsulating Network Layer data (IP, IPX, etc.) in frames

- Authentication of communicating devices
- IP address assignment for network communication
- Support for multiple network layer protocols simultaneously

PPP Frame :

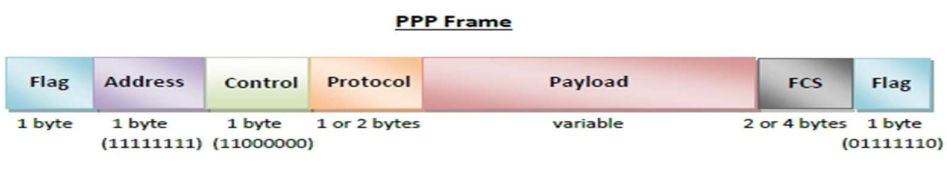


fig 22 : PPP Data Frame

Field	Size	Description
Flag	1 byte	Start/end delimiter: 01111110
Address	1 byte	Set to 11111111 (broadcast — all stations)
Control	1 byte	Set to constant 11000000
Protocol	1 or 2 bytes	Identifies the Network Layer protocol in the payload
Payload	Variable (max 1500 bytes)	Data from the Network Layer
FCS	2 or 4 bytes	CRC for error detection

Components of PPP :

- Encapsulation Component: Encapsulates datagrams so they can be sent over the physical layer.
- Link Control Protocol (LCP): Responsible for establishing, configuring, testing, maintaining, and terminating links. Negotiates options between endpoints.
- Network Control Protocol (NCP): Configures the Network Layer protocols being used over the link.

- Authentication Protocols: PPP supports two authentication methods:
  - Password Authentication Protocol (PAP): Simple two-way handshake. Password sent in clear text — less secure.
  - Challenge Handshake Authentication Protocol (CHAP): Three-way handshake using a challenge-response mechanism. More secure.

Comparison of HDLC and PPP :

Feature	HDLC	PPP
Orientation	Bit-oriented	Byte-oriented
Link type	Point-to-point & multipoint	Point-to-point only
Protocol support	Single protocol	Multiple protocols (IP, IPX, etc.)
Authentication	Not built-in	PAP and CHAP
Standard	ISO standard	IETF RFC 1661

3.4. Logical link control (LLC) and media access control (MAC) sublayers

The Data Link Layer is divided into two sublayers by IEEE 802 standards:

- Logical Link Control (LLC) — Upper sublayer
- Media Access Control (MAC) — Lower sublayer

Logical Link Control (LLC) :

The LLC sublayer provides a uniform interface to the Network Layer, regardless of the underlying MAC method. It adds header, trailer, and acknowledgement information to frames.

Three Service Options of LLC :

- Unacknowledged Connectionless Service (Type 1): Frames are sent independently without any connection setup or acknowledgement. Suitable when the link is highly reliable (e.g., Ethernet LAN). Fast but unreliable.

- Acknowledged Connectionless Service (Type 2): No logical connection is set up, but each frame is individually acknowledged. Useful for unreliable channels such as wireless links.
- Acknowledged Connection-Oriented Service (Type 3): A logical connection is established before data transfer. Each frame is numbered and guaranteed to be received. Slowest but most reliable.

Media Access Control (MAC) :

The MAC sublayer controls how devices on a network gain access to the shared medium and permission to transmit data. It is the lower sublayer of the Data Link Layer.

MAC Addressing :

- A unique 6-byte (48-bit) address permanently burned into the Network Interface Card (NIC)
- Also called Ethernet address, hardware address, physical address, or PHY address
- First 3 bytes: Organizationally Unique Identifier (OUI) — identifies the manufacturer
- Last 3 bytes: Unique identifier assigned by the manufacturer for that specific card.
- Written in hexadecimal, e.g., 00:1A:2B:3C:4D:5E

Functions of MAC Sublayer :

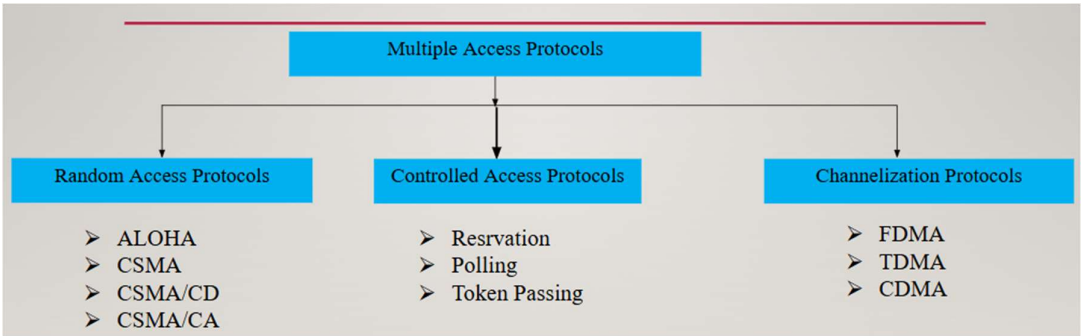
- Provides an abstraction of the physical layer to the LLC and upper OSI layers
- Encapsulates frames to make them suitable for transmission via the physical medium
- Resolves the addressing of source and destination stations
- Performs multiple access resolution — determines which station can transmit on the shared medium
- Performs collision resolution and initiates retransmission in case of collisions
- Generates the Frame Check Sequence (FCS) to protect against transmission errors

Media Access Methods :

- Contention-based: ALOHA, CSMA, CSMA/CD, CSMA/CA
- Polling: Primary station polls each secondary station in turn.

- Token Passing: A special control frame (token) is passed around; only the station holding the token may transmit.

3.5. Random access protocols: ALOHA, CSMA, CSMA/CD, CSMA/CA



In random access protocols, no station is superior to another, and no fixed schedule exists for transmission. Any station can transmit when it has data, depending on the state of the medium.

- No fixed time slot or sequence for transmission
- Multiple stations may attempt to transmit simultaneously, leading to collisions
- Collision resolution mechanisms are employed

ALOHA :

ALOHA was the first random access protocol, originally designed for wireless LAN at the University of Hawaii. It forms the basis for all later contention protocols.

Pure ALOHA :

When a station sends data it waits for an acknowledgement. If the acknowledgement doesn't come within the allotted time, then the station waits for a random amount of time called back-off time (Tb) and re-sends the data. Since different stations wait for different amount of time, the probability of further collision decreases.

Slotted ALOHA :

It is like pure aloha, except that we divide time into slots and sending of data is allowed only at the beginning of these slots. If a station misses out the allowed time, it must wait for the next slot. This reduces the probability of collision.

CSMA (Carrier Sense Multiple Access) :

CSMA improves on ALOHA by requiring a station to listen to (sense) the medium before transmitting. If the channel is busy, the station defers its transmission.

CSMA Persistence Methods :

- 1-Persistent CSMA: If channel is idle, transmit immediately with probability 1. If busy, keep sensing and transmit as soon as channel becomes idle. High collision probability when multiple stations are waiting.
- Non-Persistent CSMA: If channel is busy, wait a random time before sensing again. Reduces collision probability but wastes channel capacity during random wait.
- p-Persistent CSMA: If channel is idle, transmit with probability p. Defer with probability (1-p) to the next slot. Balances between the above two extremes.

CSMA/CD (Carrier Sense Multiple Access with Collision Detection) :

CSMA/CD adds collision detection to CSMA. A station monitors the channel during its own transmission. If a collision is detected, it immediately stops transmission and takes corrective action. Used in wired Ethernet (IEEE 802.3).

CSMA/CD Procedure :

1. Sense the channel. If idle, start transmitting.
2. Continue sensing while transmitting.
3. If a collision is detected, send a jam signal to notify all stations.
4. Stop transmitting. Wait for a random back-off time (Binary Exponential Backoff).
5. Try again from step 1.

Binary Exponential Backoff: After each collision, the maximum random wait time is doubled (up to 10 times, then capped). After 16 failed attempts, the transmission is aborted.

CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance) :

CSMA/CA tries to avoid collisions before they occur rather than detect them after the fact. It is used in wireless networks (IEEE 802.11 Wi-Fi) where collision detection is impractical.

CSMA/CA Procedure :

1. Sense the channel. If idle for a time interval called DIFS (Distributed Inter-Frame Space), proceed.

2. Choose a random back-off time from a contention window. Count down only while channel is idle.
3. When back-off expires, transmit the frame.
4. Wait for ACK. If received, success. If not, assume collision and double the contention window, try again.

RTS/CTS mechanism: Optional — sender sends Request to Send (RTS), receiver replies with Clear to Send (CTS), reserving the channel and solving the hidden node problem.

3.6. Token based protocols: Token bus, Token ring, FDDI

Token Bus :

Token Bus uses a physical bus topology but creates a logical ring among stations for token passing. Stations are ordered in a logical sequence and the token passes from one station to the next in that order.

- Used primarily in industrial and manufacturing environments (MAP - Manufacturing Automation Protocol)
- A station that receives the token may transmit. After it's done (or has no data), it passes the token to the next logical station
- More complex to implement than Token Ring

Advantage: Deterministic access time - suitable for real-time industrial control.

Token Ring :

In Token Ring, all stations are connected in a logical ring topology. A token (small frame) continuously circulates around the ring. A station that wants to transmit must wait to capture the free token.

Operation :

1) A free token circulates around the ring.
2) A station needing to transmit captures the token and changes it to a busy token.
3) The station appends its data and sends the frame around the ring.
4) The destination station copies the data and marks the frame as received.
5) When the frame returns to the sender, the sender removes it and releases a new free token.

Speed: 4 Mbps or 16 Mbps.

Advantage: No collisions, deterministic, fair access.

Disadvantage: If any station or the token is lost, the ring fails.

FDDI (Fiber Distributed Data Interface) :

FDDI is a token-passing protocol designed for high-speed fibre-optic networks. It uses a dual-ring architecture for fault tolerance.

- Uses two counter-rotating rings: a primary ring and a secondary ring
- Normally, data travels on the primary ring. If the primary ring fails, data is automatically rerouted onto the secondary ring, creating a single larger ring
- Speed: 100 Mbps
- Distance: Up to 200 km total ring length
- Supports up to 500 stations
- Used as a backbone network (MAN/WAN) in the 1990s, largely superseded by Fast Ethernet and Gigabit Ethernet

Feature	Token Ring (802.5)	Token Bus (802.4)	FDDI
Topology	Physical ring	Physical bus, logical ring	Dual counter-rotating ring
Speed	4 or 16 Mbps	1, 5, or 10 Mbps	100 Mbps
Medium	Twisted pair/STP	Coaxial cable	Fiber optic
Fault tolerance	Monitor station	Limited	Dual ring (high)
Primary use	Office LANs	Industrial networks	Backbone MAN/WAN

3.7. Ethernet (IEEE 802.3) and its evolution, bridged and switched Ethernet

Ethernet is the most widely deployed LAN technology today. It uses CSMA/CD for medium access control and was standardized by IEEE as 802.3.

Traditional Ethernet (10BASE-T) :

- Speed: 10 Mbps
- Uses CSMA/CD for access control

- Frame size: 64 to 1518 bytes
- Topology: Originally bus, then star (with hub at centre)

Ethernet Frame Structure :

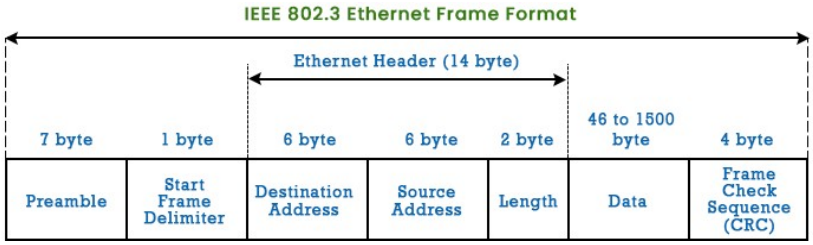


fig 23: Ethernet Frame Structure

Field	Size	Description
Preamble	7 bytes	Synchronization pattern (10101010...)
SFD (Start Frame Delimiter)	1 byte	10101011 - marks start of frame
Destination Address	6 bytes	MAC address of receiver
Source Address	6 bytes	MAC address of sender
Length/Type	2 bytes	Length of data or EtherType (protocol ID)
Data + Padding	46–1500 bytes	Payload; padded if less than 46 bytes
FCS	4 bytes	CRC-32 for error detection

Evolution of Ethernet :

Standard	Speed	Medium	IEEE Standard
Standard Ethernet	10 Mbps	Coax / UTP / Fiber	IEEE 802.3
Fast Ethernet	100 Mbps	UTP Cat5 / Fiber	IEEE 802.3u
Gigabit Ethernet	1 Gbps	UTP Cat5e/6 / Fiber	IEEE 802.3ab/z

Standard	Speed	Medium	IEEE Standard
10-Gigabit Ethernet	10 Gbps	Fiber / UTP Cat6a	IEEE 802.3ae
100-Gigabit Ethernet	100 Gbps	Fiber	IEEE 802.3ba

Bridged Ethernet :

A bridge operates at the Data Link Layer and connects two or more LAN segments. It filters traffic based on MAC addresses to reduce collisions and separate collision domains.

- A bridge is a repeater with added functionality of filtering by MAC address
- Has a single input and output port - effectively a 2-port device
- Learns MAC addresses by observing traffic and builds a forwarding table

Types of Bridges :

- Transparent Bridges: These are the bridge in which the stations are completely unaware of the bridge's existence i.e. whether a bridge is added or deleted from the network, reconfiguration of the stations is unnecessary. These bridges make use of two processes i.e. bridge forwarding and bridge learning.
- Source Routing Bridges : In these bridges, routing operation is performed by source station, and the frame specifies which route to follow. The host can discover frame by sending a special frame called discovery frame, which spreads through the entire network using all possible paths to destination.

Switched Ethernet :

A switch is a multiport bridge. Instead of sharing the medium among all stations, each station gets its own dedicated connection to the switch, eliminating collisions.

- Each port on a switch is its own collision domain
- Operates at the Data Link Layer
- Performs error checking before forwarding frames - only forwards valid frames
- Dramatically increases available bandwidth compared to shared (hub-based) Ethernet
- Full-duplex communication possible between each station and the switch

Feature	Hub (Shared Ethernet)	Switch (Switched Ethernet)
OSI Layer	Physical Layer (Layer 1)	Data Link Layer (Layer 2)
Collision Domain	One for all ports	One per port
Bandwidth sharing	Shared among all ports	Dedicated per port
Duplex	Half-duplex	Full-duplex possible
Intelligence	None — broadcasts all frames	Forwards based on MAC address table

3.8. WLAN (Wi-Fi) and IEEE 802.11

A Wireless Local Area Network (WLAN) allows devices to connect without physical cables, using radio waves. The IEEE 802.11 family of standards governs WLANs, commonly known as Wi-Fi.

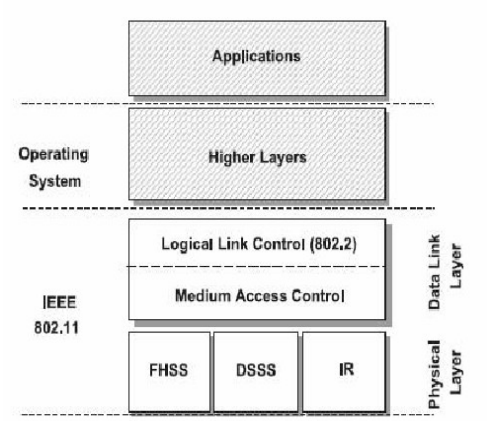


fig 24 : IEEE 802.11 protocol stack

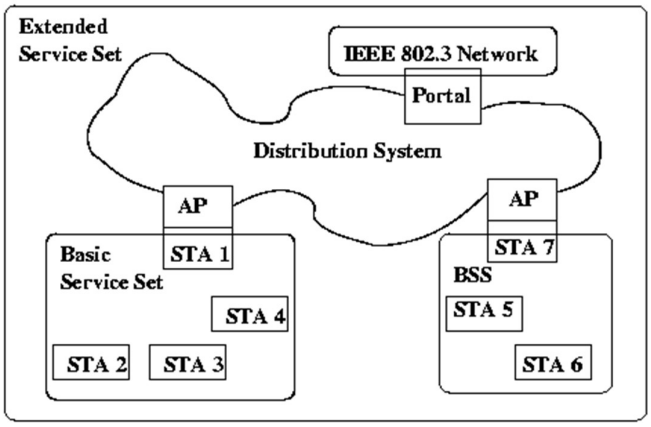


fig 25 : IEEE 802.11 architecture

Components :

- Basic Service Set (BSS): The fundamental building block of a WLAN. Contains a group of stations communicating with each other.
- Access Point (AP): A device that acts as the coordinator for a BSS. It connects the wireless network to the wired network.
- Extended Service Set (ESS): Two or more BSSs connected via a Distribution System (DS) - typically a wired Ethernet backbone.
- IBSS (Ad Hoc mode): A BSS without an AP - stations communicate directly with each other.

MAC Layer in IEEE 802.11 :

IEEE 802.11 uses CSMA/CA (not CSMA/CD) because:

- Collision detection requires a station to transmit and receive simultaneously — impractical at the power levels used in wireless LANs
- Hidden node problem: Two stations may not be able to hear each other, but both can reach the AP
- CSMA/CA avoids collisions using random back-off and optional RTS/CTS handshake

IEEE 802.11 Standards Evolution :

Standard	Year	Frequency	Max Speed	Notes
802.11	1997	2.4 GHz	2 Mbps	Original standard
802.11b	1999	2.4 GHz	11 Mbps	First widely adopted Wi-Fi
802.11a	1999	5 GHz	54 Mbps	Less interference, shorter range
802.11g	2003	2.4 GHz	54 Mbps	Backward compatible with 802.11b
802.11n	2009	2.4/5 GHz	600 Mbps	Wi-Fi 4; uses MIMO
802.11ac	2013	5 GHz	6.93 Gbps	Wi-Fi 5; MU-MIMO
802.11ax	2021	2.4/5/6 GHz	9.6 Gbps	Wi-Fi 6; OFDMA, BSS coloring

Hidden Node and Exposed Node Problems :

- Hidden Node Problem: Station A and C are both in range of AP B, but not in range of each other. A cannot detect C's transmission, causing a collision at B. Solved by RTS/CTS mechanism.
- Exposed Node Problem: A station refrains from transmitting because it senses a busy medium even though its own transmission would not cause interference.

3.9. VLAN and its importance

A Virtual Local Area Network (VLAN) is a logical grouping of network devices within the same broadcast domain, regardless of their physical location. VLANs are created by software configuration on switches.

How VLANs Work ?

- A managed switch can logically divide its ports into separate VLANs.
- Devices in the same VLAN can communicate as if they were on the same physical LAN.

- Devices in different VLANs cannot communicate directly — they require a router or Layer 3 switch.
- VLAN membership is typically based on port number, MAC address, or protocol.

Importance of VLANs :

- Broadcast Control: VLANs limit the scope of broadcast traffic. Broadcasts are confined to the VLAN rather than the entire physical LAN, reducing congestion.
- Security: Sensitive departments (Finance, HR) can be isolated in their own VLAN, preventing unauthorized access from other network segments.
- Flexibility and Scalability: Users can be moved between VLANs by software reconfiguration rather than physical rewiring. New VLANs can be created as the organization grows.
- Performance: Reducing broadcast domains reduces unnecessary traffic, improving overall network performance.
- Cost Reduction: Multiple logical networks can run over a single physical infrastructure, reducing the need for separate hardware.

Types of VLANs :

- Port-based VLAN: Each switch port is assigned to a VLAN. Most common method.
- MAC-based VLAN: Devices are assigned to VLANs based on their MAC address. More flexible - follows the device.
- Protocol-based VLAN: Traffic is assigned to VLANs based on the Layer 3 protocol (e.g., IP, IPX).

3.10. Virtual circuit switching: ATM, MPLS

Virtual circuit switching sets up a logical path (virtual circuit) between source and destination before data is transferred. All packets follow the same path and arrive in order. It combines characteristics of both circuit switching and packet switching.

Virtual Circuit Concepts :

- Virtual Circuit (VC): A logical connection between source and destination. Resources are not dedicated (unlike physical circuit switching), but a path is established.

- Virtual Circuit Identifier (VCI): A label that identifies the virtual circuit. Routers/switches use VCI to forward packets along the pre-established path.
- Setup phase: A control message establishes the virtual circuit and sets up forwarding tables at each node.
- Data transfer phase: Data is sent along the established path using the VCI.
- Teardown phase: The virtual circuit is released after communication ends.

ATM (Asynchronous Transfer Mode) :

ATM is a high-speed networking standard designed for both LAN and WAN that uses fixed-size cells (53 bytes each) to transfer data, voice, and video over the same network.

ATM Cell Structure :

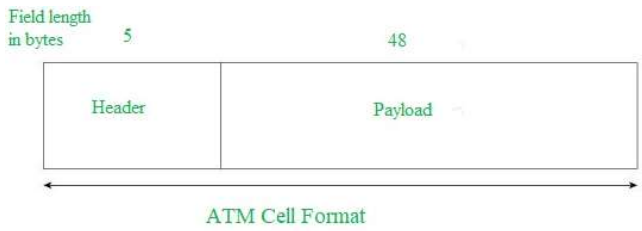


fig 26 : ATM cell structure

Field	Size	Description
Header	5 bytes	Contains VPI (Virtual Path Identifier), VCI, and control info
Payload	48 bytes	User data
Total cell size	53 bytes	Fixed - enables hardware-based switching at high speeds

ATM Connection Hierarchy :

- Virtual Path (VP): A bundle of virtual circuits between the same two endpoints.
- Virtual Channel (VC): An individual logical connection within a virtual path.

ATM Characteristics :

- Connection-oriented: A virtual circuit is established before data transmission

- Fixed cell size: 53 bytes. Header = 5 bytes, payload = 48 bytes
- Speed: Originally designed for 155 Mbps to 622 Mbps, can scale to Gbps
- Supports QoS (Quality of Service) - can guarantee bandwidth, delay, and jitter for voice/video
- Categories: CBR (Constant Bit Rate), VBR (Variable Bit Rate), ABR (Available Bit Rate), UBR (Unspecified Bit Rate)

MPLS — Multiprotocol Label Switching :

MPLS is a data forwarding mechanism that uses short fixed-length labels (rather than long IP addresses) to make forwarding decisions. It operates between the Data Link Layer and Network Layer - sometimes called a Layer 2.5 protocol.

How MPLS Works :

1. An ingress Label Edge Router (LER) receives a packet and assigns a label based on the destination and traffic class.
2. The label is pushed (added) to the front of the packet.
3. Interior Label Switching Routers (LSRs) forward the packet based solely on the label — no IP lookup needed.
4. Each LSR swaps the incoming label with an outgoing label (label swapping).
5. At the egress LER, the label is popped (removed) and the original packet is forwarded normally.

MPLS Label Stack Entry (32 bits) :

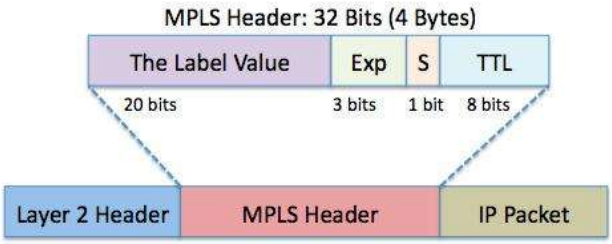


fig 27 : MPLS Stack

Field	Bits	Description
Label value	20 bits	The forwarding label
Traffic Class (TC)	3 bits	QoS / Differentiated Services Code Point
Stack bit (S)	1 bit	1 if this is the bottom label in the stack
TTL	8 bits	Time to Live (decremented at each hop)

Advantages of MPLS :

- Faster packet forwarding - label lookup is faster than IP address longest prefix match.
- Traffic Engineering - control over specific paths for load balancing and QoS.
- Multiprotocol support - works with IP, ATM, Ethernet, Frame Relay.
- Enables VPNs (MPLS VPNs) at service provider scale.
- Supports QoS - different traffic classes (voice, video, data) can be handled differently.

Comparison of ATM and MPLS :

Feature	ATM	MPLS
Cell/packet size	Fixed (53 bytes)	Variable (uses existing frames)
Layer	Data Link / Physical	Between Layer 2 and 3 (Layer 2.5)
Identifier	VPI/VCI	Label (20-bit)
Protocol support	ATM-specific	Multiprotocol
QoS	Strong built-in QoS	Supports DiffServ-based QoS
Deployment	Carrier backbone (legacy)	Widely used in ISP cores